Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Advanced Java Programming

BCA — Semester 6 — 2024 — Model Answer Sheet

Subject Code: CACS354

Note: This is a model answer sheet for study purposes. Answers are based on standard curriculum topics.

Group B

Attempt any SIX questions. [6×5=30]

2. What is AWT? Differentiate between AWT and Swing with examples.

Answer: AWT (Abstract Window Toolkit): Java's original GUI framework. Heavyweight components (use OS native resources). Limited components, platform-dependent look. **Swing:** Built on AWT, lightweight components (rendered in Java). Richer component set, pluggable look-and-feel, platform-independent. **Differences:** 1) AWT components are heavyweight; Swing are lightweight. 2) Swing has more components (JTable, JTree, JTabbedPane). 3) Swing supports MVC architecture. 4) AWT has platform-dependent look; Swing is consistent. 5) Swing requires javax.swing package.

3. Explain the MVC architecture. How is it implemented in Java web applications?

Answer: MVC (Model-View-Controller): Design pattern separating concerns. **Model:** Business logic and data (JavaBeans, POJOs, database entities). **View:** User interface (JSP, HTML, Thymeleaf). **Controller:** Handles requests, processes input, selects views (Servlets, Spring Controller). **Implementation in Java:** Servlet acts as Controller, receives HTTP requests, calls Model methods, forwards to JSP View. Frameworks like Spring MVC formalize this pattern with annotations (@Controller, @RequestMapping, @ModelAttribute).

4. What is JDBC? Explain the steps to connect a Java application to a database using JDBC.

Answer: JDBC (Java Database Connectivity): API for connecting Java applications to databases. **Steps:** 1) Load driver: Class.forName('com.mysql.cj.jdbc.Driver'). 2) Establish connection: DriverManager.getConnection(url, user, password). 3) Create statement: Connection.createStatement() or prepareStatement(). 4) Execute query: executeQuery() for SELECT, executeUpdate() for INSERT/UPDATE/DELETE. 5) Process results: Iterate through ResultSet. 6) Close resources: Close ResultSet, Statement, Connection. **Types:** Statement, PreparedStatement (precompiled, prevents SQL injection), CallableStatement (stored procedures).

5. Define servlet. Explain the servlet lifecycle with a diagram.

Answer: A **Servlet** is a Java class that handles HTTP requests and generates responses. It runs on a web server (Tomcat, Jetty). **Lifecycle:** 1) **Loading:** Web container loads servlet class. 2) **Instantiation:** Creates servlet object. 3) **init():** Called once for initialization. 4) **service():** Called for each request. Dispatches to doGet(), doPost(), etc. 5) **destroy():** Called once before removing servlet. Releases resources. **Advantages:** Platform-independent, efficient (persistent), powerful (full Java API), secure.

6. What is JSP? Differentiate between JSP and Servlet. Explain JSP lifecycle.

Answer: JSP (Java Server Pages): Technology for creating dynamic web content using HTML with embedded Java code. **JSP vs Servlet:** JSP is presentation-focused (HTML with Java); Servlet is logic-focused (Java with HTML). JSP is easier for UI developers; Servlet is better for complex logic. JSP is translated to Servlet internally. **JSP Lifecycle:** 1) Translation: JSP converted to Servlet source. 2) Compilation: Servlet compiled to class. 3) Loading/Instantiation: Class loaded. 4) init(): _jspInit() called. 5) service(): _jspService() handles requests. 6) destroy(): _jspDestroy() for cleanup.

7. Explain Java event handling model with a program to handle button click events.

Answer: Java Event Handling: Based on Delegation Event Model. **Components: Event Source:** Component generating events (button). **Event Object:** Contains event info (ActionEvent). **Event Listener:** Interface implementing handler methods (ActionListener). **Process:** 1) Create event source (JButton). 2) Create listener implementing ActionListener. 3) Register listener with source using addActionListener(). 4) Implement actionPerformed() method. **Example:** JButton btn = new JButton('Click'); btn.addActionListener(e -> JOptionPane.showMessageDialog(null, 'Button clicked!'));

8. What is JavaBean? Explain the properties and advantages of using JavaBeans.

Answer: A **JavaBean** is a reusable software component following conventions: public no-arg constructor, private properties with getters/setters, implements Serializable. **Properties:** Simple (single value), Indexed (arrays), Bound (notifies listeners on change), Constrained (listeners can veto changes). **Advantages:** Reusability across applications, portability (platform-independent), compact and easy to create, supports introspection (development tools can discover properties), can be used in visual builders, supports persistence. **Usage:** In JSP with , in Spring as managed beans.

Group C

Attempt any TWO questions. [2×10=20]

9. Explain RMI (Remote Method Invocation) architecture. Write a program to implement a simple calculator using RMI.

Answer: RMI Architecture: Enables Java objects on different JVMs to invoke methods on each other. **Layers: Stub/Skeleton Layer:** Stub on client side marshals calls; skeleton on server unmarshals. **Remote Reference Layer:** Manages references to remote objects. **Transport Layer:** Handles actual network communication. **Components:** Remote interface (declares remote methods), Remote object (implements interface), Registry (binds remote objects to names), Client (looks up and invokes). **Calculator Example:** 1) Define Remote interface extending java.rmi.Remote with methods like add(int, int). 2) Implement interface extending UnicastRemoteObject. 3) Create server, bind to Registry using Naming.rebind(). 4) Client uses Naming.lookup() to get remote reference. 5) Call methods as if local. **Security:** Requires SecurityManager and policy file.

10. What is multithreading? Explain thread lifecycle and synchronization in Java. Write a program to demonstrate thread synchronization.

Answer: Multithreading: Executing multiple threads concurrently within a single process. **Thread Lifecycle:** New → Runnable → Running → Waiting/Timed Waiting → Blocked → Terminated. **Creation:** Extend Thread or implement Runnable. **Synchronization:** Prevents race conditions when multiple threads access shared data. **Methods: synchronized keyword:** On methods or blocks. **wait()/notify():** For inter-thread communication. **ReentrantLock:** More flexible locking. **Example:** class Counter { synchronized void increment() { count++; } }. Multiple threads calling increment() are serialized, ensuring count is accurate. **Deadlock:** Can occur if threads wait for each other's locks. **Best Practices:** Minimize lock scope, use concurrent collections, avoid nested locks.

11. Explain JSP implicit objects and standard actions. Write a JSP program to display employee details from a database using JDBC.

Answer: JSP Implicit Objects: Predefined variables available in JSP without declaration. **request:** HttpServletRequest. **response:** HttpServletResponse. **session:** HttpSession. **application:** ServletContext. **out:** JspWriter for output. **config:** ServletConfig. **pageContext:** PageContext. **page:** current servlet instance. **exception:** Throwable (error pages). **Standard Actions:** (instantiate bean), (set property), (get property), (include resource), (forward request). **Employee Program:** Use , , then JDBC code in scriptlet or JSTL to fetch and display records in HTML table.

These model answers are prepared for educational purposes. Refer to textbooks for comprehensive study.